

Commutation

COMMUTATION

For the purpose of power control or power conditioning a conducting thyristor must be turned off as desired.

Turn off implies bringing a thyristor from forward conducting state to forward blocking state.

When a thyristor is turned off, the current carried by it has to be transferred to another part of the circuit.

This process of turning off a thyristor is termed commutation.

The thyristor turn-off requires the fulfillment of following conditions:

- (1) Anode current should fall below holding current.
- (2) ^{Preferably,} A reverse voltage should be applied to the thyristor [^]for a sufficient time to enable it to recover to blocking state.

Commutation is broadly classified under two categories:-²³

(1) NATURAL COMMUTATION

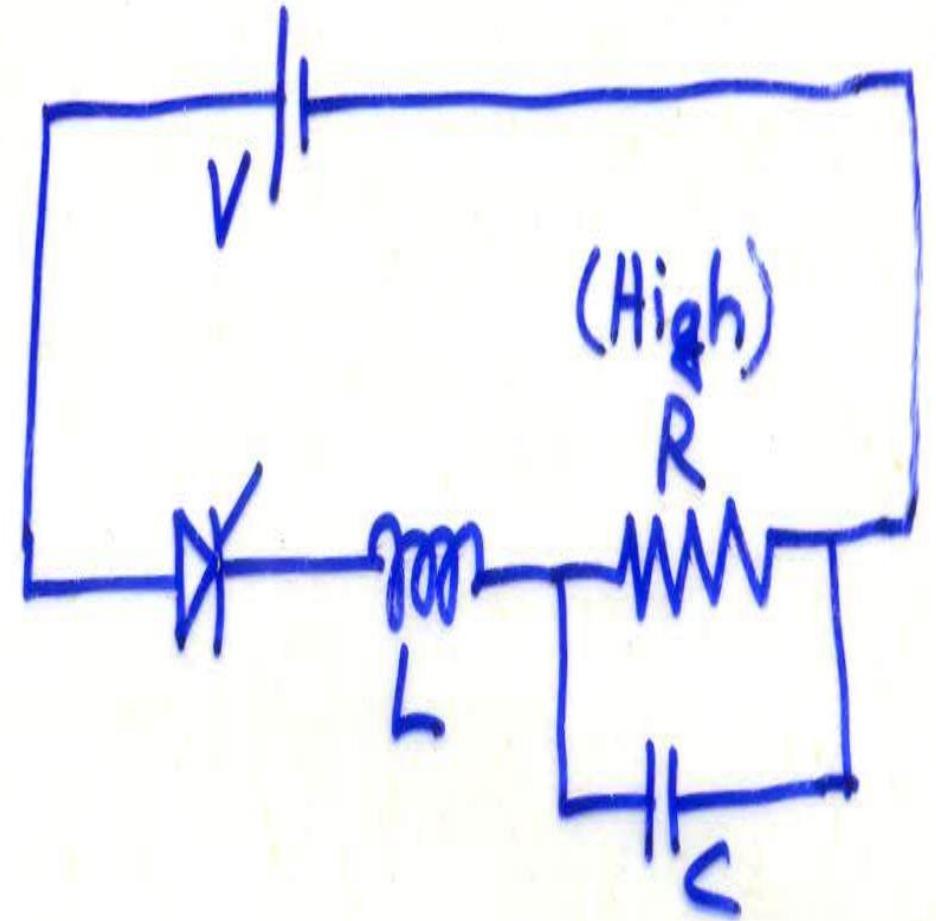
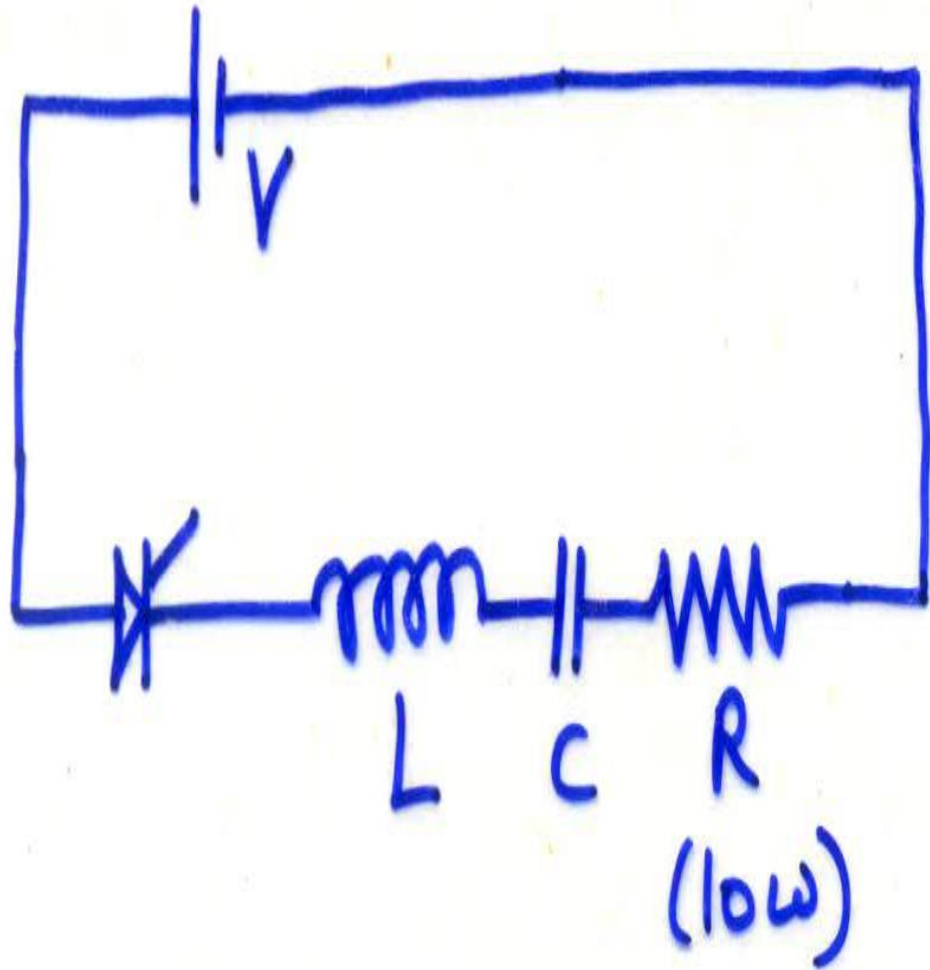
When an SCR is connected to an AC source, the current goes through its natural 0 at the end of every half cycle & a reverse voltage appears automatically across the SCR. This phenomenon would turn off the thyristor if the turn off time of thyristor is less than the duration of the -ve half cycle. This is known as natural commutation $\therefore$ no external or special circuit is required to turn off the thyristor.

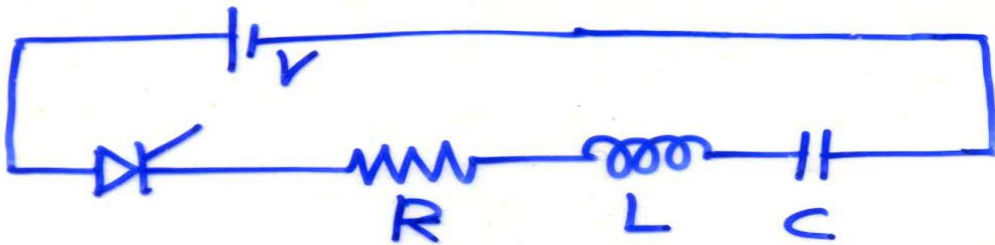
(2) FORCED COMMUTATION

In DC circuits there is no natural 0 of current. In such circuits SCR have to be turned off by a special commutation circuit using extra components. This process is known as forced commutation.

The inductors & capacitors used to achieve forced commutation are called commutating components.

(1) Resonant / Self / Load Commutation





$$\frac{V}{s} = R I(s) + sL I(s) + \frac{1}{sC} I(s)$$

$$\Rightarrow \frac{I(s)}{V} = \frac{1}{\left[R + sL + \frac{1}{sC} \right] s}$$

$$\Rightarrow \text{Characteristic Eq. } Ls^2 + Rs + \frac{1}{C} = 0$$

$$\text{or } s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$$

$$\text{Compare with } s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$\Rightarrow 2\zeta\omega_n = \frac{R}{L} \dots (i) \quad \omega_n = \frac{1}{\sqrt{LC}} \dots (ii)$$

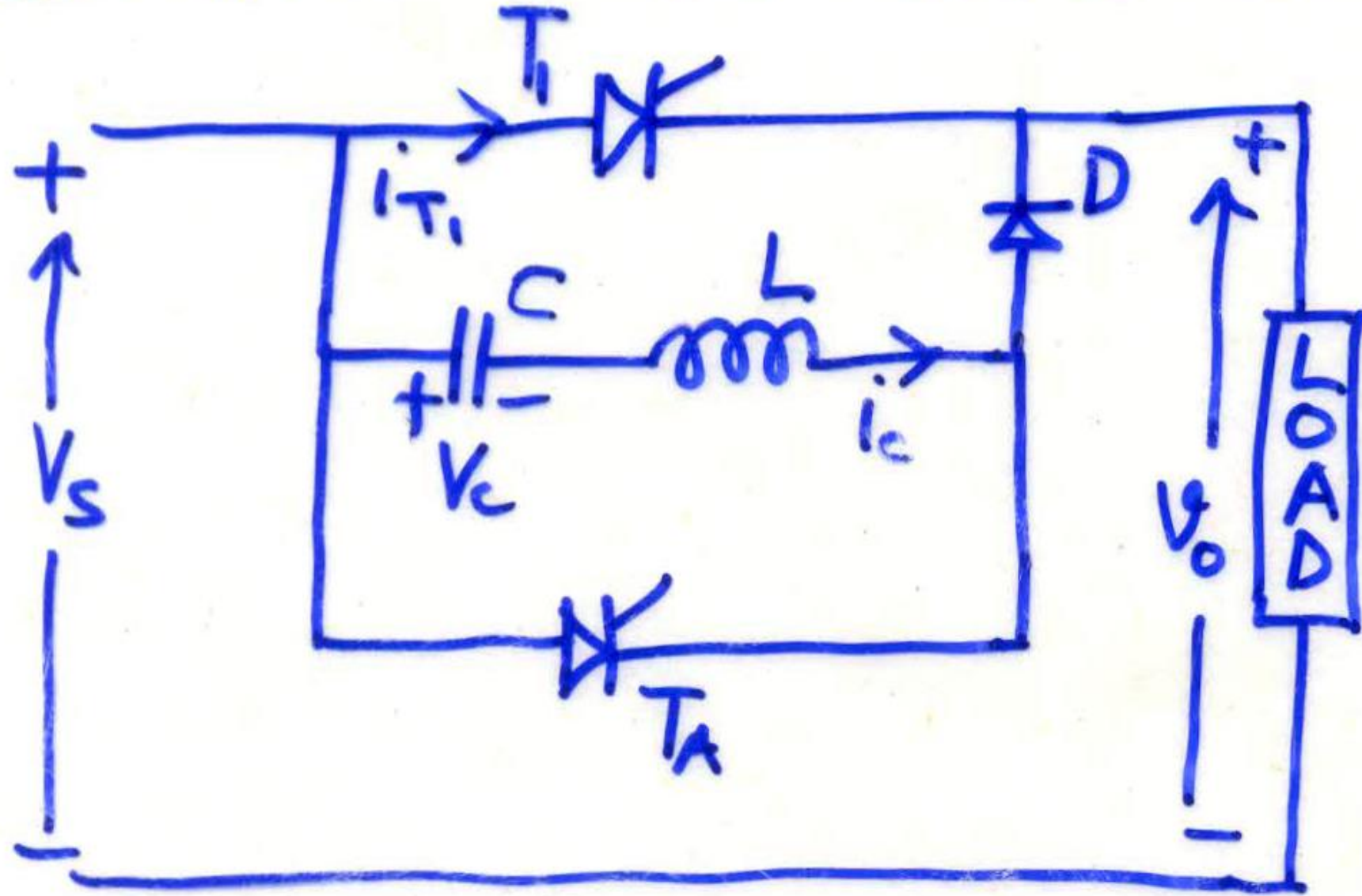
$$\Rightarrow \omega_d = \omega_n [1 - \zeta^2]^{1/2} = \left[\frac{4L - R^2C}{4L} \right]^{1/2} \frac{1}{\sqrt{LC}}$$

For underdamped system $\omega_d > 0$

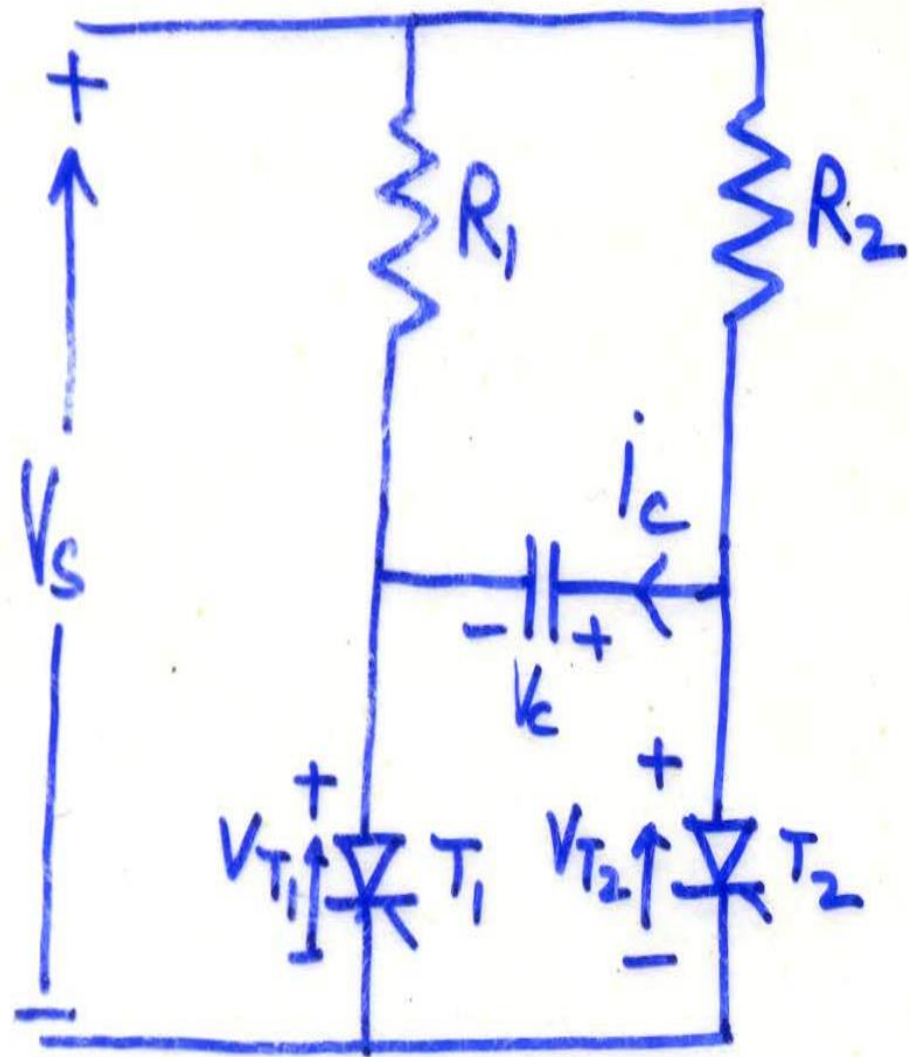
$$\Rightarrow \frac{1}{LC} - \frac{R^2}{4L^2} > 0$$

$$\Rightarrow R < \left[\frac{4L}{C} \right]^{1/2}$$

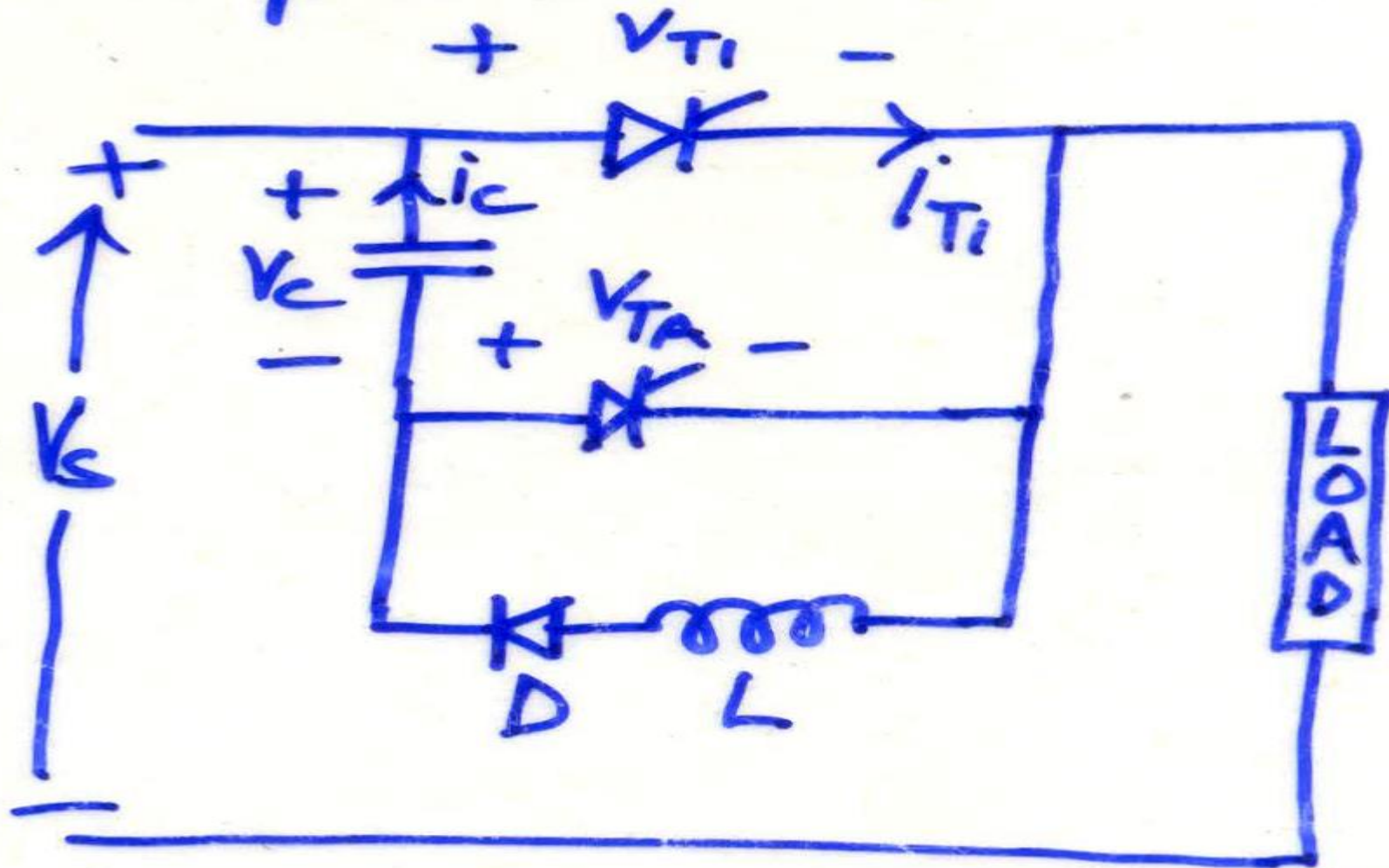
(2) Resonant Pulse Commutation



(B) Complementary / Complementary Impulse Commutation



(4) Impulse/Auxiliary/Parallel Capacitor Commutation



5) External Pulse Commutation

